

In [66]:

```
#import relevant modules
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from sympy import *
from scipy import stats
from datetime import timedelta
import seaborn as sns
```

In [2]:

```
#read data into notebook
data = 'D:\AI Course\Applied Machine Intelligence\Covid vaccination types.csv'
```

In [3]:

```
#read the data into the notebook
Vac = pd.read_csv(data)
```

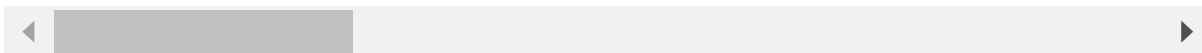
In [4]:

Vac

Out[4]:

	country	iso_code	date	total_vaccinations	people_vaccinated	people_fully_vac
0	Afghanistan	AFG	22/02/2021	0.0	0.0	
1	Afghanistan	AFG	23/02/2021	NaN	NaN	
2	Afghanistan	AFG	24/02/2021	NaN	NaN	
3	Afghanistan	AFG	25/02/2021	NaN	NaN	
4	Afghanistan	AFG	26/02/2021	NaN	NaN	
...	...	...	...	...	...	
86507	Zimbabwe	ZWE	25/03/2022	8691642.0	4814582.0	34
86508	Zimbabwe	ZWE	26/03/2022	8791728.0	4886242.0	34
86509	Zimbabwe	ZWE	27/03/2022	8845039.0	4918147.0	34
86510	Zimbabwe	ZWE	28/03/2022	8934360.0	4975433.0	35
86511	Zimbabwe	ZWE	29/03/2022	9039729.0	5053114.0	35

86512 rows × 15 columns



In [5]:

Vac['vaccines'].nunique()

Out[5]:

84

In [6]:

```
#remove unneeded columns
Vac.drop(columns=['source_name', 'source_website', 'iso_code', 'daily_vaccinations_raw', 'dail
```

In [8]:

```
Vac.dropna(inplace=True)
```

In [9]:

```
Vac
```

Out[9]:

	country	date	total_vaccinations	people_vaccinated	people_fully_vaccinated	d
78	Afghanistan	11/05/2021	504502.0	448878.0	55624.0	
87	Afghanistan	20/05/2021	547901.0	470341.0	77560.0	
91	Afghanistan	24/05/2021	573277.0	476367.0	96910.0	
93	Afghanistan	26/05/2021	590454.0	479372.0	111082.0	
94	Afghanistan	27/05/2021	593313.0	479574.0	113739.0	
...	...	...	...	...	...	
86507	Zimbabwe	25/03/2022	8691642.0	4814582.0	3473523.0	
86508	Zimbabwe	26/03/2022	8791728.0	4886242.0	3487962.0	
86509	Zimbabwe	27/03/2022	8845039.0	4918147.0	3493763.0	
86510	Zimbabwe	28/03/2022	8934360.0	4975433.0	3501493.0	
86511	Zimbabwe	29/03/2022	9039729.0	5053114.0	3510256.0	

37406 rows × 10 columns

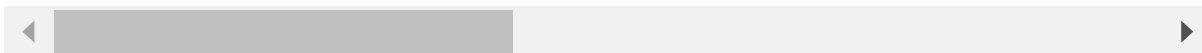
In [11]:

```
Vac[Vac['vaccines'].str.contains('Oxford/AstraZeneca')]
```

Out[11]:

	country	date	total_vaccinations	people_vaccinated	people_fully_vaccinated	d
78	Afghanistan	11/05/2021	504502.0	448878.0	55624.0	
87	Afghanistan	20/05/2021	547901.0	470341.0	77560.0	
91	Afghanistan	24/05/2021	573277.0	476367.0	96910.0	
93	Afghanistan	26/05/2021	590454.0	479372.0	111082.0	
94	Afghanistan	27/05/2021	593313.0	479574.0	113739.0	
...	...	...	...	...	...	
86507	Zimbabwe	25/03/2022	8691642.0	4814582.0	3473523.0	
86508	Zimbabwe	26/03/2022	8791728.0	4886242.0	3487962.0	
86509	Zimbabwe	27/03/2022	8845039.0	4918147.0	3493763.0	
86510	Zimbabwe	28/03/2022	8934360.0	4975433.0	3501493.0	
86511	Zimbabwe	29/03/2022	9039729.0	5053114.0	3510256.0	

29843 rows × 10 columns



In [13]:

```
Vac[Vac['vaccines'].str.contains('Pfizer')]
```

Out[13]:

ns	total_vaccinations_per_hundred	people_vaccinated_per_hundred	people_fully_vaccinated_per_h
1.0	1.27		1.13
2.0	1.38		1.18
2.0	1.44		1.20
3.0	1.48		1.20
7.0	1.49		1.20
...	...		...
5.0	218.33		79.71
7.0	218.40		79.72
3.0	218.41		79.72
5.0	218.46		79.75
3.0	218.53		79.77

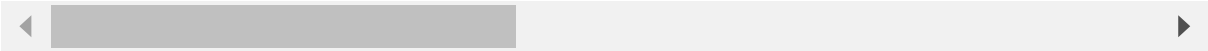


In [14]:

```
#only look for countires that only have one vaccine type in their country
display(Vac[Vac["vaccines"]=="Oxford/AstraZeneca"])
```

	country	date	total_vaccinations	people_vaccinated	people_fully_vaccinated	daily
1723	Angola	12/05/2021	626572.0	586377.0	40195.0	
1729	Angola	18/05/2021	681502.0	606287.0	75215.0	
1732	Angola	21/05/2021	757535.0	630984.0	126551.0	
1737	Angola	26/05/2021	859979.0	663049.0	196930.0	
1742	Angola	31/05/2021	909215.0	679034.0	230181.0	
...	...	...	...	...	...	
83777	Vanuatu	20/12/2021	152711.0	102308.0	50403.0	
83812	Vanuatu	24/01/2022	178464.0	109176.0	69288.0	
83826	Vanuatu	07/02/2022	184045.0	110235.0	73810.0	
83840	Vanuatu	21/02/2022	189138.0	110907.0	78231.0	
83847	Vanuatu	28/02/2022	199464.0	114284.0	85180.0	

531 rows × 10 columns

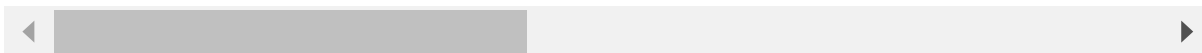


In [16]:

```
display(Vac[Vac["vaccines"]=="Pfizer/BioNTech"])
```

	country	date	total_vaccinations	people_vaccinated	people_fully_vaccinated	daily
3664	Aruba	10/04/2021	41096.0	29197.0	11899.0	
3666	Aruba	12/04/2021	43073.0	31139.0	11934.0	
3667	Aruba	13/04/2021	45588.0	33563.0	12025.0	
3668	Aruba	14/04/2021	48200.0	36058.0	12142.0	
3669	Aruba	15/04/2021	50546.0	37981.0	12565.0	
...	...	...	...	...	...	
80510	Turks and Caicos Islands	11/02/2022	67463.0	30801.0	28811.0	
80517	Turks and Caicos Islands	18/02/2022	67965.0	31082.0	29032.0	
80524	Turks and Caicos Islands	25/02/2022	68176.0	31165.0	29160.0	
80538	Turks and Caicos Islands	11/03/2022	68464.0	31296.0	29317.0	
80552	Turks and Caicos Islands	25/03/2022	68670.0	31419.0	29400.0	

398 rows × 10 columns

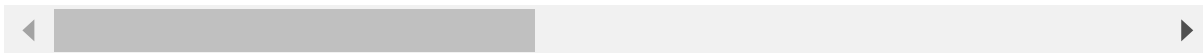


In [17]:

```
display(Vac[Vac["vaccines"]=="Moderna"])
```

	country	date	total_vaccinations	people_vaccinated	people_fully_vaccinated	dai
30726	Greenland	01/03/2021	5130.0	3927.0	1203.0	
30765	Greenland	09/04/2021	9555.0	5296.0	4259.0	
30768	Greenland	12/04/2021	9808.0	5531.0	4277.0	
30769	Greenland	13/04/2021	9996.0	5700.0	4296.0	
30770	Greenland	14/04/2021	10142.0	5845.0	4297.0	
...	...	...	...	...	...	
85326	Wallis and Futuna	30/11/2021	11939.0	6016.0	5923.0	
85368	Wallis and Futuna	11/01/2022	12247.0	6130.0	6117.0	
85375	Wallis and Futuna	18/01/2022	12287.0	6151.0	6136.0	
85416	Wallis and Futuna	28/02/2022	12849.0	6450.0	6399.0	
85437	Wallis and Futuna	21/03/2022	12940.0	6483.0	6457.0	

162 rows × 10 columns



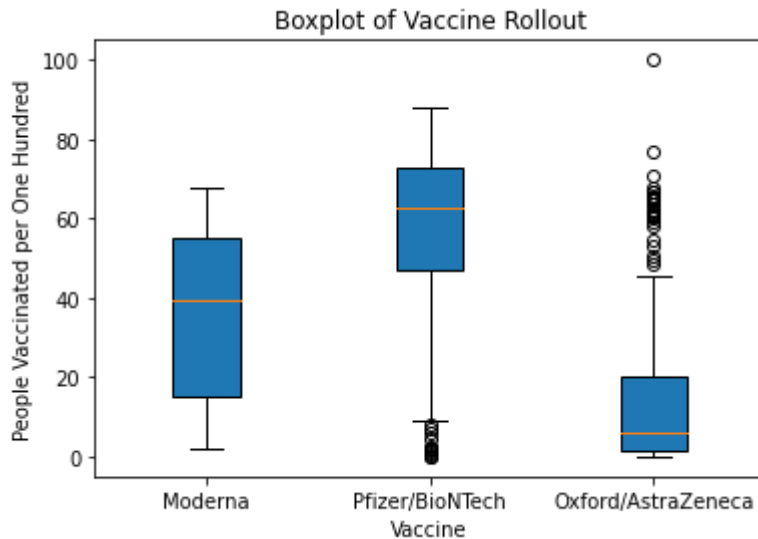
In [20]:

```
#create array fro the three daatsets
Moderna_vac = Vac[Vac["vaccines"]=="Moderna"].people_fully_vaccinated_per_hundred
Pfizer_vac = Vac[Vac["vaccines"]=="Pfizer/BioNTech"].people_fully_vaccinated_per_hundred
Oxford_vac = Vac[Vac["vaccines"]=="Oxford/AstraZeneca"].people_fully_vaccinated_per_hundred
```

In [24]:

```
#create the box plot
box_plot_data=[Moderna_vac,Pfizer_vac,Oxford_vac]
plt.boxplot(box_plot_data,patch_artist=True,labels=['Moderna','Pfizer/BioNTech','Oxford/Ast

plt.xlabel('Vaccine')
plt.ylabel('People Vaccinated per One Hundred')
plt.title('Boxplot of Vaccine Rollout')
plt.show()
```

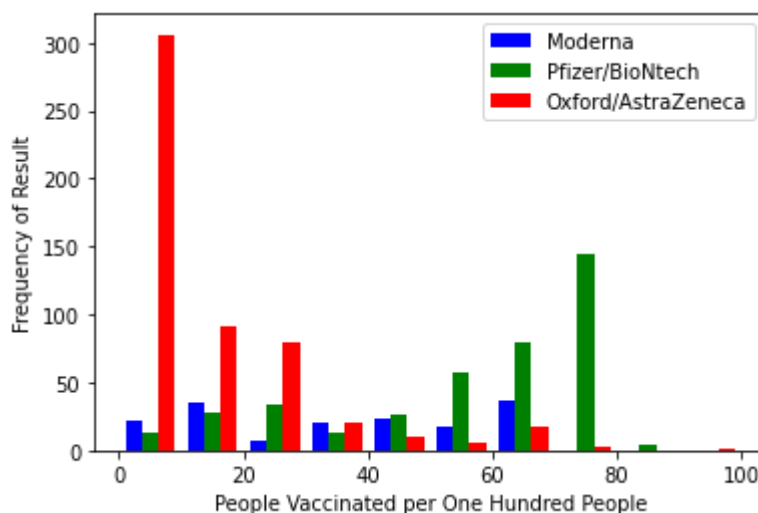


In [63]:

```
#create the histogram
x = Moderna_vac
y = Pfizer_vac
z = Oxford_vac

plt.hist([x,y,z],color=['b','g','r'], label=['Moderna','Pfizer/BioNtech','Oxford/AstraZeneca
plt.ylabel("Frequency of Result")
plt.xlabel("People Vaccinated per One Hundred People")
plt.legend(loc='upper right')

plt.show()
```



In [64]:

```
#f test
anova = stats.f_oneway(Moderna_vac, Pfizer_vac, Oxford_vac)

print(anova, "\n")
```

F_onewayResult(statistic=594.3014006310916, pvalue=3.646249898398261e-175)

In [65]:

```
#test to see if there is any significance
alpha = 0.05

if(anova[1] < alpha):
    print("Significant difference exists between which vaccine we choose")
else:
    print("There is no significant difference in which vaccine we choose")
```

Significant difference exists between which vaccine we choose

In []: